

Detailed Outline - Midterm

All the material, comments, etc. in the slides.

I. EXPONENTIAL AND POISSON

PDF in resources:

- Exponential distribution (inter arrival, service) + Poisson process
- Properties: memoryless (+ proof), minimum interarrival time of multiple exponential processes (distribution, average, probability of arrival from process i), aggregate of exponential processes, sub-sampled exponential process.

II. ARCHITECTURE

Tanenbaum:

- 2.6.1: Telephone network topology
- 2.6.2: LATA, LEC, IXC, POP, definitions and description
- 2.6.4: trunks, multiplexing, hierarchy of multiplexing, TDM, FDM. **No** details on digitalization and modulation (Only comments in slides), **No** SONET/SDH, **No** wavelength multiplexing
- 2.6.5: circuit switching

Kurose:

- 1.3.2: circuit switching, multiplexing in circuit switching
- 1.2.1: access networks
- 1.3.1: Packet switching
- 1.3.2: Packet switching vs circuit switching
- 1.3.3: Network of networks
- 1.4: performance metrics: delay, throughput, packet loss, (efficiency in slides)
- 1.5: layering: motivation, ISO stack, internet stack, encapsulation
- ~~3.6.1: congestion and scenarios.~~

III. APPLICATION LAYER

Kurose:

- 2.1: client-server and peer-to-peer architectures, processes and sockets, transport services, TCP and UDP services, Application layer protocols
- 2.2: architecture, non-persistent and persistent connections (no header format), cookies
- 2.6: architecture and motivation, bitTorrent (architecture and operations).
- 7.1: whole sub-chapter
- 7.2: 7.2.1, 7.2.2
- 7.3: best-effort and QoS (packet loss, delay, jitter), Removing jitter, Recovering packet loss (FEC, interleaving)
- 7.4: RTP (basics, fields)